

SYED MUHAMMAD ALI GILANI

Electrical Engineer | Electronics | PEC Registered Engineer

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PROFESSIONAL SUMMARY

Electrical Engineering graduate specializing in Electronics with experience in embedded systems, robotics workshops, sensor integration, and Arduino-based system development. Strong academic foundation in electronic circuit design, microcontrollers and hardware integration. Familiar with GNSS/INS navigation systems, circuit simulation and technical documentation. Seeking an entry-level Electronics, Embedded Systems, or Hardware Engineer position.

EDUCATION

Bachelor of Electrical Engineering

Federal Urdu University of Arts, Science and Technology (FUUAST), Islamabad

CGPA: 3.22/4.00

PROFESSIONAL EXPERIENCE

Power Systems Trainee – 32 MW Hydropower Project

- Gained practical exposure to hydropower generation processes and plant operations.
- Developed understanding of turbine, synchronous generator, excitation and governor systems.
- Learned Emergency Shutdown (ESD) procedures and Lockout/Tagout (LOTO) practices

Transformer Testing & Diagnostics Trainee – Hammad Engineering & Transformer Co.

- Developed understanding of transformer oil testing, dielectric strength, moisture analysis and DGA.
- Reviewed transformer condition assessment reports and preventive maintenance practices.
- Assisted in diagnostic procedures used to improve equipment reliability.

CORE COMPETENCIES

- Embedded Systems
- Electronic Circuit Design
- Analog & Digital Electronics
- Microcontroller Programming
- Sensor Integration
- Robotics & Automation
- Hardware Troubleshooting
- Technical Documentation

TECHNICAL SKILLS

- Engineering Software: MATLAB, Simulink, Proteus, AutoCAD, Arduino IDE, Microsoft Office
 - Industrial exposure to electrical systems complemented by electronics-focused academic projects and robotics activities.
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FINAL YEAR PROJECT

GNSS/INS-Based Vehicle Navigation Prototype

- Designed a prototype vehicle integrating Arduino Uno, GPS and IMU sensors for basic navigation.
 - Integrated hardware components and developed navigation logic using Arduino IDE.
 - Performed calibration, testing, troubleshooting and technical documentation.
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PROFESSIONAL DEVELOPMENT & LEADERSHIP

- Organizer – YOUTH ROBOTECH
 - Participant – AI INNOFEST (Bahria University)
 - Participant – National Youth Conference (PICSS)
 - Workshops: GNSS/INS Navigation, Line Follower Robot, Home Automation, RC Plane
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LANGUAGES

English | Urdu